Harshith Kantamneni

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SUMMARY

AI systems engineer specializing in **multi-agent orchestration**, **tool-use integration**, and production deployment of **LLM-powered applications**. Designs and operates three **fully autonomous agent organizations** built on the **Anthropic Claude API**, including **MCP server** design, **sub-agent architectures**, **three-tier memory systems**, and independent **evaluation frameworks**. GPU performance engineering background (CUDA, Triton, gem5) provides deep understanding of the inference stack underneath agentic workloads.

TECHNICAL SKILLS

Agent Systems: Multi-agent orchestration, tool-use integration, MCP (Model Context Protocol) server design, sub-agent architectures, agent loops, context management, multi-step reasoning, agent evaluation frameworks, advanced prompt engineering, structured outputs, checkpoint/resume, agent observability, attestation verification

AI / ML APIs & SDKs: Anthropic Claude API, Claude Code, Claude Agent SDK, OpenAI API, ElevenLabs, fal.ai, LLM function calling, RAG patterns, sentence-transformers

Languages: Python, TypeScript, JavaScript, Rust, C17, CUDA, Triton, Metal

Infrastructure: Supabase (Postgres / Auth / Storage), Railway, tmux session persistence, stream-JSON processing, SQLite FTS5, ripgrep, cargo workspaces + cargo-deny, Docker, REST APIs, CI/CD (pytest, vitest, ruff, eslint, tsc), Git

GPU & Inference: NVIDIA Nsight Compute / Systems, CUDA kernel optimization, Triton autotuning, GPU memory hierarchy, roofline modeling, gem5 simulation, LLM inference profiling

PROJECTS

AGI Research Lab: Autonomous Multi-Agent Research Organization | *Mar 2026 - Present*

Python, Anthropic Claude API, Claude Code, tmux, stream-JSON, Anthropic memory_20250818 protocol

- Built a **fully autonomous research lab** of **30 agent roles across 8 hierarchical layers** running `claude-opus-4-7` and pursuing scientific questions through a **15-phase program flow**; PI/Director co-equal governance requires **unanimous-compromise protocols** on 7 load-bearing decision classes, mediated by a dedicated sub-agent.
- **Detected and remediated a document-forgery incident in production**: Director agent fabricated 5 formal signatures across 6 sessions; root cause was cost-asymmetry between agent dispatch and attestation. Built a **three-layer structural fix** (signature verifier cross-referencing episodic dispatch logs, runner-level fail-closed exit code, and mandatory **independent Evaluator** check) that blocks graceful session exit on detection.
- Engineered the session runner as a state machine handling 7 exit conditions (graceful checkpoint, context-full, rate-limit, evaluator-fail, forgery-unaddressed, victory, catastrophic); implemented **three-tier memory architecture** (hot 40 KB / wiki 50 KB / log 30 KB rolling) via Anthropic's `memory_20250818` protocol with archive-never-delete invariant.

HIVE Product Lab: Autonomous Multi-Agent Product Company | *Mar 2026 - Present*

Rust (cargo workspaces), Python, Claude Code, SQLite FTS5, ripgrep, MCP, cargo-deny

- Operates a **fully autonomous product-company agent organization** (38 specialist roles spanning product, engineering, design, legal, strategy, and ops) shipping a Rust reasoning engine + P2P node through formal alpha gates; **1,688+ tests, zero failures** at the gate.
- Designed **scoped-packet agent dispatch** where agents receive specific file paths and line ranges, are forbidden from wholesale state reads, and return 200-word summaries with detail on disk. Cut per-cycle startup **context from ~100K to ~15-20K tokens**.
- Built an in-house memory CLI with **dual backend** (filesystem source-of-truth + SQLite FTS5 search accelerator) supporting semantic search, permanent IDs across archival, and cross-reference resolution (**116/116 tests passing**); architected independent **Evaluator agent** running 19+ discipline checks per cycle against raw stream-JSON to verify model compliance and PI-directive coverage.

The Obsidian Archive: Autonomous Multi-Agent Content Pipeline | *Feb 2026 - Present*

Python, Anthropic Claude API, ElevenLabs, Epidemic Sound (MCP), fal.ai, Supabase, Railway

- Architected a **15-stage autonomous pipeline with 13 specialized agents** for end-to-end YouTube documentary production (topic discovery, research, fact verification, scripting, compliance review, TTS, video assembly, upload); runs unattended on Railway cron.
- Built **checkpoint/resume** on Supabase-backed state (any stage recovers from failure without re-executing priors), series-continuity detection that propagates parent context across multi-part content, and a **MIRROR feedback agent** that closes the loop by writing retention-derived optimization signals back to upstream agents.
- Integrated 5+ external APIs, including **Epidemic Sound via the MCP protocol**, with rate limiting, retry logic, and graceful degradation. Extended the pipeline with a companion **110-agent hierarchical evaluation framework** (structured-debate subsystem, adversarial red team, convergence gates) that reviews every output before publication.

EDUCATION

University of Wisconsin-Madison | Master of Science, Electrical and Computer Engineering | *December 2025*

Coursework: Computer Architecture (grad), Machine Learning, HPC with CUDA, Fault-Tolerant Computing

GPU Portfolio: 76 LLaMA/Mistral GEMM shapes on A100 with **1.73× Triton-vs-cuBLAS speedup** via fusion; **XGBoost + NN CUDA kernel autotuner** within 5% of exhaustive search; co-authored gem5 study of AMD MI300X memory hierarchy.